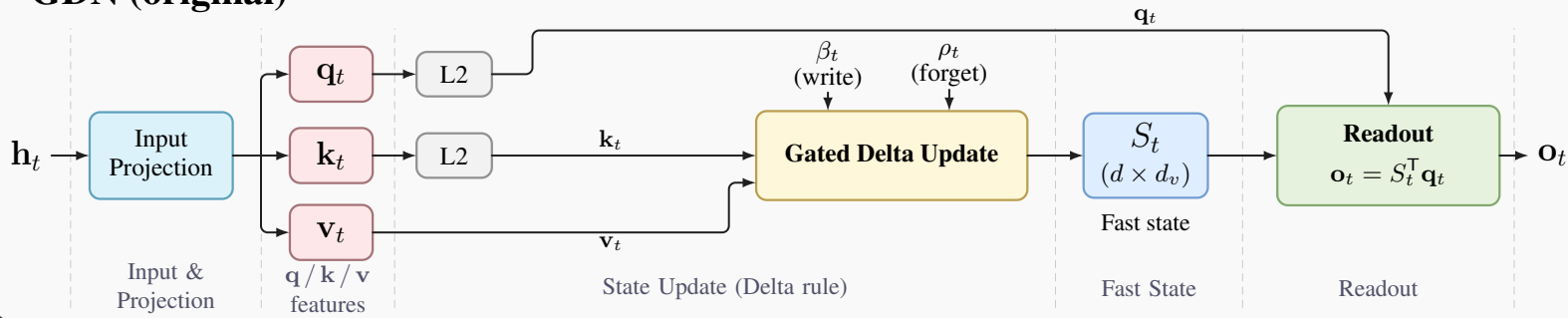
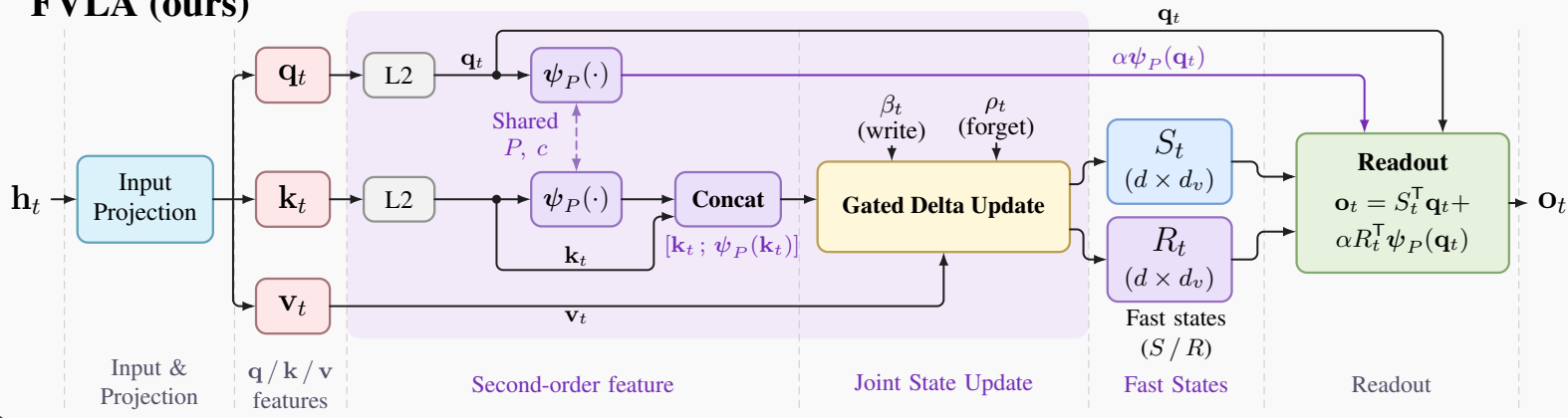


GDN (original)



FVLA (ours)



Linear layer



Feature ($\mathbf{q}$ / $\mathbf{k}$ / $\mathbf{v}$)



Normalization



Second-order feature



State update (Delta rule)



Fast state ($\mathbf{S}$ / $\mathbf{R}$)



Readout

$\psi_P(\mathbf{x}) = c [\mathbf{x} \odot (P\mathbf{x})]$